Interlude on tensor products (Section 6.1). The left-hand side of the equation is an element of a 4 × 4 matrix. Each of its two indices can take four distinct values: *uu*, *ud*, *du*, or *dd*. What about the right-hand side? The matrix element $L_{a'a}$ also has two indices, but each of them can take only two distinct values: *u* or *d*. In fact, the same symbol *L* refers to two different matrices on each side of Eq. 7.18.

At first glance, it appears as though we have equated a 4 × 4 matrix to a 2 × 2 matrix, and indeed that would be a problem. However, the factor $\delta_{b'b}$ makes everything work out. The term $L_{a'a}\ \delta_{b'b}$ is an element of the tensor product of two 2 × 2 matrices, and that tensor product *is* a 4 × 4 matrix.[6] Here is the way to read Eq. 7.18:

> *The* 4 × 4 *matrix* $L_{a'b',ab}$ *can be factored into a tensor product of the two* 2 × 2 *matrices* $L_{a'a}$ *and* $\delta_{b'b}$*, where* $\delta_{b'b}$ *is equivalent to the* 2 × 2 *identity matrix.*

Now, let's calculate the expectation value of **L** (the 4 × 4 version) using the full apparatus of the composite system:

$$\langle\Psi|\mathbf{L}|\Psi\rangle = \sum_{a,b,a',b'} \psi^*(a',b')\ L_{a'b',ab}\ \psi(a,b).$$

As I warned, there are lots of indices. But it gets simpler if we use the special form of the matrix *L*. The factor $\delta_{b'b}$ in Eq. 7.18—a Kronecker delta—filters out any elements that change Bob's half of the label, and leaves the others intact. It tells us to set $b' = b$ to get

$$\langle\Psi|\mathbf{L}|\Psi\rangle = \sum_{a',b,a} \psi^*(a',b)\ L_{a',a}\ \psi(a,b). \qquad (7.19)$$

For the moment, let's ignore the sums over *a* and *a′*, and concentrate instead on the sum over *b*. We encounter the quantity